

Differentiation Write Up

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1 Definitions

Definition 1. We define

$$\lim_{x \rightarrow c} f(x) = L \quad (1)$$

if $\forall \epsilon > 0, \exists \delta > 0$ s.t. $|x - c| < \delta \implies |f(x) - L| < \epsilon$.

Definition 2. Let f be defined (and real-valued) on $[a, b]$. For any $x \in [a, b]$ form the quotient

$$\phi(t) = \frac{f(t) - f(x)}{t - x} \quad (a < t < b, t \neq x) \quad (2)$$

and define

$$f'(x) = \lim_{t \rightarrow x} \phi(t) \quad (3)$$

provided the limit exists as per definition 1.

2 Theorems

Theorem 1. Suppose $f'(x)$, $g'(x)$ exist, $g'(x) \neq 0$ and $f(x) = g(x) = 0$. Prove that

$$\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)}. \quad (4)$$

Proof. Since $f(x) = g(x) = 0$ it follows that $\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{g(t) - g(x)} = \lim_{t \rightarrow x} \frac{\frac{f(t) - f(x)}{t - x}}{\frac{g(t) - g(x)}{t - x}}$. Since $f'(x)$ and $g'(x)$ exist and $g'(x) \neq 0$, then by Definition 2 and the Algebraic Limit Theorem, $\lim_{t \rightarrow x} \frac{\frac{f(t) - f(x)}{t - x}}{\frac{g(t) - g(x)}{t - x}} = \frac{\lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x}}{\lim_{t \rightarrow x} \frac{g(t) - g(x)}{t - x}} = \frac{f'(x)}{g'(x)}$ as desired. $\square$

Theorem 2. Let f be defined for all real x , and suppose that

$$|f(x) - f(y)| \leq (x - y)^2 \quad (5)$$

for all real x and y . Then $f(x)$ is constant.

Proof. Suppose that f is defined on all real x and that $|f(x) - f(y)| < (x - y)^2$ for all real x and y . Then $|f(x) - f(y)| \leq (x - y)^2 = |x - y|^2 \implies \frac{|f(x) - f(y)|}{|x - y|} \leq |x - y|$. Taking limit as $x \rightarrow y$ yields

$$\lim_{x \rightarrow y} y - x \leq \lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y} \leq \lim_{x \rightarrow y} x - y \quad (6)$$

$$0 \leq \lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y} \leq 0 \quad (7)$$

$$f'(x) = \lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y} = 0 \quad (8)$$

So $f'(x) = 0$ for all real x (note that this assumed $x \geq y$ if that was not the case we just flip the inequalities in (6) and (7) but the result remains the same). Therefore $f(x)$ is constant. $\square$

Theorem 3. Assume $f : (0, 2) \rightarrow \mathbb{R}$ is differentiable and there is a constant $M > 0$ so that $|f'(x)| \leq M$ for all $x \in (0, 2)$. Let $a_n = f(1/n)$ for $n \in \mathbb{N}$. Prove that (a_n) is a convergent sequence.

Proof. Since f is differentiable on $(0, 2)$ it follows that f is continuous on $(0, 2)$ and thus by Mean Value Theorem, for any $n, m \in (0, 2)$ it follows that there exists a constant $c \in (\frac{1}{2\max\{m, n\}}, 1)$ (we cannot guarantee that f is continuous at 0) such that $f'(c) = \frac{f(1/n) - f(1/m)}{(1/n) - (1/m)}$. Thus $|f(1/n) - f(1/m)| = |f'(c)| |(1/n) - (1/m)| \leq M |1/n - 1/m|$. Since $(1/n)$ is convergent (and thus Cauchy) it follows that $\forall \epsilon > 0, \exists N > 0$ s.t. $n, m > N \implies |1/n - 1/m| < \frac{\epsilon}{M}$. Thus $n, m > N \implies |f(1/n) - f(1/m)| \leq M |1/n - 1/m| < M(\frac{\epsilon}{M}) = \epsilon$. Therefore (a_n) is Cauchy and thus convergent. $\square$